ConvNet Size & Parameters

Simple · STUDENT — fill in the blanks

One black-and-white image through a small ConvNet — the lecture example.

THE ONLY TWO FORMULAS (stride 1, no padding)

```
Conv2D:   params = ((M x N x L) + B) x F       new size = old size - (M - 1)
MaxPool:  params = 0                           new size = old size / pool
Flatten:  params = 0                           size = rows x cols x channels
Dense:    params = (inputs + 1) x nodes        size = nodes
```

M, N = kernel size   L = channels coming IN (inherited, never chosen)
B = bias = 1   F = filters = feature maps coming OUT (you choose)

THE CODE (given)

```
model = models.Sequential()
model.add(layers.Conv2D(3, (5, 5), activation='relu',
                        input_shape=(28, 28, 1)))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(5, (3, 3), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Flatten())
model.add(layers.Dense(256, activation='relu'))
model.add(layers.Dense(2, activation='softmax'))
```

Layer	Output shape	Params	The math
Conv2D: F=3, (5x5)			
MaxPool 2x2			
Conv2D: F=5, (3x3)			
MaxPool 2x2			
Flatten			
Dense(256)			
Dense(2, softmax)			
TOTAL trainable parameters			

Fill Output shape + Params for each layer, then total. Check against model.summary().